PRODUCT REQUIREMENT DOCUMENT

# PRD — Live Stock & Options Trading Scanner

PRD-2026 · AIAvenue Agency

# Executive Summary

India's equity derivatives segment on the NSE sees over ₹70,000 Cr in daily F&O turnover, driven by 2.5+ million active retail traders. The vast majority of these traders rely on scattered tools — a price chart for technicals, a scanner for volume, Excel for OI data, and Twitter for market sentiment — creating cognitive overload and missed opportunities. This product is a real-time NSE F&O analysis platform that scans all 188 F&O equities through a 7-filter Pro Scanner pipeline (Market Trend, Sector Strength, PDH/PDL Breakout, Dynamic Volume/ATR, VWAP, RSI 14, OI Buildup), an AI Intraday Scanner with a 7-factor scoring engine, and a 4-Phase Smart Scanner that drills from Market Macro → Sector Heatmap → PDH/PDL → Deep Analytics. Additionally, it provides a NIFTY OI tracker with PCR, support/resistance levels, and a multi-factor sentiment engine. All data streams in real time via Socket.io. Target TAM is ₹250 Cr, SAM ₹62.5 Cr, SOM ₹12.5 Cr in the Indian retail trading tools market.

---

# Problem Statement

## Core Problem

Retail F&O traders in India operate in an information-asymmetric environment. Institutional traders have Bloomberg Terminals, Reuters Eikon, and dedicated quantitative research teams. Retail traders have TradingView (limited NSE data), broker-provided tools (basic, ad-heavy, 15-min delayed), and WhatsApp groups (noise-heavy, unreliable). The specific pain points cluster around four gaps:

1. **Data Fragmentation** — Price, OI, volume, sector data, market mood — all in different tools 2. **No Contextual Scanning** — Existing scanners show raw numbers; they don't rank or contextualise 3. **Time Pressure** — F&O decisions must be made in seconds; manual cross-referencing is impossible at scale 4. **Pattern Blindness** — Traders miss macro-to-micro connections (e.g., banking sector strength → specific stock PDH breakout)

## User Pain Points

| Persona | Pain Point | Frequency | Severity |
|---|---|---|---|
| Retail F&O Trader | Spends 45 mins/day manually screening 188 stocks across 3 platforms | Daily | Critical |
| Swing Trader | Misses breakout setups because he can't watch 40 stocks simultaneously | Intraday | High |
| Options Writer | Cannot accurately gauge sector-wide OI buildup for correlated hedging | Weekly | High |
| Trading Educators | Need to demonstrate professional-grade scanning to students but have no tool to show | Per class | Medium |
| Proprietary Desk | Junior analysts spend 2+ hours pre-market preparing scan reports | Daily | Critical |

## Core Problem

Retail F&O traders in India operate in an information-asymmetric environment. Institutional traders have Bloomberg Terminals, Reuters Eikon, and dedicated quantitative research teams. Retail traders have TradingView (limited NSE data), broker-provided tools (basic, ad-heavy, 15-min delayed), and WhatsApp groups (noise-heavy, unreliable). The specific pain points cluster around four gaps:

1. **Data Fragmentation** — Price, OI, volume, sector data, market mood — all in different tools 2. **No Contextual Scanning** — Existing scanners show raw numbers; they don't rank or contextualise 3. **Time Pressure** — F&O decisions must be made in seconds; manual cross-referencing is impossible at scale 4. **Pattern Blindness** — Traders miss macro-to-micro connections (e.g., banking sector strength → specific stock PDH breakout)

## User Pain Points

---

# User Research & Personas

## Persona 1: Karan Malhotra — Retail F&O Trader

| Attribute | Detail |
|---|---|
| Age | 34 |
| Location | Gurugram |
| Day Job | Senior Software Engineer (trades 2–3 hours daily) |
| Portfolio | ₹12 Lakh capital; trades NIFTY Bank, Reliance, Infy, HDFC Bank options |
| Tools Used | TradingView charts + Zerodha's Kite (broker) + Sensibull (options analysis) + Twitter screener |
| Frustration | "I have 4 tabs open. By the time I piece together volume spike + OI buildup + VWAP crossover on a stock, the move is already done. I need one screen that connects the dots." |
| Need | Unified scanner with multi-factor ranking; real-time alerts; OI + price divergence detection |

## Persona 1: Karan Malhotra — Retail F&O Trader

## Persona 2: Meera Iyengar — Swing Trader / Educator

## Persona 3: Ajay Suri — Proprietary Trading Desk Analyst

---

# Market Analysis

## Total Addressable Market (TAM)

Indian retail trading tools and analytics market (2026): **₹500 Cr**

- 2.5 million active F&O traders
- Average annual spend on tools: ₹2,000–₹10,000/trader
- Plus institutional/prop desk spend

F&O-specific tools segment: **50%** of the above.

**TAM = ₹500 Cr × 50% = ₹250 Cr**

## Serviceable Addressable Market (SAM)

Traders who actively use at least 2 paid tools (TradingView + broker/platform) and trade at least 10 lots/month. Estimated **25%** of the active F&O base.

**SAM = ₹250 Cr × 25% = ₹62.5 Cr**

## Serviceable Obtainable Market (SOM)

Target 10,000 paid users + 50 prop desk/institution seats over 3 years:

- Retail: 10,000 users × ₹6,000/year = ₹6 Cr
- Institutional: 50 seats × ₹1,00,000/year = ₹0.5 Cr
- Base Year 3 → scale to **₹12.5 Cr** by Year 5

**SOM = ₹12.5 Cr (5-year target)**

## Market Rationale

- SEBI has tightened F&O regulations (2025–26), increasing the need for professional-grade tools
- Zerodha's trading volume dropped 30% post-SEBI circular; traders need an edge more than ever
- No web-based scanner exists that combines price + OI + VWAP + RSI + volume + sector + market macro in a single view
- TradingView's NSE data is not real-time (15-min delayed for free; real-time costs extra + has gaps)
- The market is fragmented across dozens of small tools; no clear winner has emerged

---

# Competitive Analysis

| Competitor | Type | Strengths | Weaknesses | Our Edge |
|---|---|---|---|---|
| **TradingView** | Charting + Scanning | Best-in-class charting; strong community; Pine Script customisation | 15-min delayed NSE data (free); no OI data; no sector heatmap; no F&O-specific filters | Real-time NSE data; native F&O OI; 7-filter Pro Scanner with F&O context |
| **Sensibull** | Options Analysis | Excellent options payoff diagrams; strategy builder | No equity scanning; no multi-factor ranking; no real-time scanner | Equity + options scanning with factor-based ranking |
| **Amibroker** | Scanning + Backtesting | Most powerful custom scanning (AFL); massive library of scans | Desktop-only (no web/mobile); dated UI; learning curve | Web-based modern UI; zero-coding scanner builder; 4-phase guided pipeline |
| **Opstra** | Options Analytics | Good OI analytics; Indian market focused | Limited to options chain data; no price-action scanning; no alerts | Full price + OI + volume + technical scanning; real-time alerts |
| **Investing.com** | Financial Data Portal | Free; broad coverage; economic calendar | Not real-time; no F&O-specific tools; ad-heavy; no custom scanning | Real-time F&O-focused scanning; ad-free; customisable |
| **Chartink** | Indian Stock Screener | Free; Indian stocks; basic technical scans | Intraday data gaps; no OI data; basic UI; limited filters | 7 professional-grade filters; OI + sector + market macro; modern UI |

**Key Differentiators:**

- **7-Filter Pro Scanner** — Market Trend → Sector Strength → PDH/PDL Breakout → Dynamic Volume/ATR → VWAP Check → RSI 14 → OI Buildup
- **AI Intraday Scanner** — 7-factor scoring engine that ranks all 188 F&O stocks in real time
- **4-Phase Smart Scanner** — Macro → Sector → PDH/PDL → Deep Analytics drill-down
- **NIFTY OI Tracker** — PCR + support/resistance + multi-factor sentiment engine
- **Socket.io** — True real-time streaming (no polling)
- **Modern Web UI** — Dark theme, responsive, saved scan configurations

---

# Solution Overview

## High-Level Architecture

```
┌──────────────────────────────────────────────────────────────┐
│ Frontend (Next.js / React) │
│ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ │ Pro │ AI │
4-Phase │ NIFTY OI │ │ │ Scanner │ Scanner │ Scanner │ Tracker + PCR │ │
│ └──────────┘ └──────────┘ └──────────┘ └──────────────────┘ │
│ ┌──────────┐ ┌──────────┐
┌──────────┐ ┌──────────────┐ │ │ Alerts │ Saved │ Export │ Settings / │ │ │
└──────────┘ └──────────┘ └──────────┘
└──────────────┘ │ (WebSocket)│ Scans │ (CSV/PDF)│ Scan Builder │ │
└──────────────────────────────────────────────────────────────┘
▲ ▲
┌──────────────────────────┬──────────┬──────────────────────────┐
│ API Layer (Node.js
+ Socket.io) │ │ ┌──────────────────────────────────────────────┐ │ │ Real-Time Engine (Socket.io) — pushed to all clients │ │ │
└──────────────────────────────────────────────┘ │ │ ┌──────────────────────────────────────────────┐ │ │ Scan Pipeline (Market Trend → Sector → PDH/PDL → ...
) │ │ │ └──────────────────────────────────────────────┘ │ │ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │ │ NSE Data │ Historical │ OI Data Aggregator │
│ │ Feed (WebSocket) │ Database │ (NSE F&O Bhavcopy) │ │ │ └──────────┘ └──────────┘ └──────────────────┘ │
└──────────────────────────────────────────────┘
```

## Scanner Pipeline Details

### 7-Filter Pro Scanner

| Filter # | Name | What It Does | Data Source |
|---|---|---|---|
| 1 | Market Trend | Determines if NIFTY/BANKNIFTY is trending up/down/sideways | NSE Spot Index |
| 2 | Sector Strength | Ranks sectors by composite momentum | NSE Sector Indices |
| 3 | PDH/PDL Breakout | Stocks breaking previous day high/low with volume confirmation | 1-min bars |
| 4 | Dynamic Volume/ATR | Volume > 1.5× 20-day avg AND ATR > 14-day ATR average | Tick data |
| 5 | VWAP Check | Price vs. VWAP (above = bullish; below = bearish) | Intraday data |

| | | | |
|---|---|---|---|
| 6 | RSI 14 | RSI between 40–70 for continuation; >70/<40 for reversal watch | 5-min bars |
| 7 | OI Buildup | OI change > 5% with price change in same direction (valid) or divergence | F&O bhavcopy |

## High-Level Architecture

```
┌──────────────────────────────────────────────────────────────┐
│ Frontend (Next.js / React) │
│ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ │ Pro │ │ AI │ │ 4-Phase │ │ NIFTY OI │ │
│ │ Scanner │ │ Scanner │ │ Scanner │ │ Tracker + PCR │ │
│ └──────────┘ └──────────┘ └──────────┘ └──────────────────┘ │
│ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ │ Alerts │ │ Saved │ │ Export │ │ Settings / │ │ (WebSocket)│ Scans │ │ (CSV/PDF)│ │ Scan Builder │ │
│ └──────────┘ └──────────┘ └──────────┘ └──────────┘ └──────────┘ └──────────┘ └──────────────────┘ │
└──────────────────────────────────────────────────────────────┘
▲ ▲
┌──────────────────────────┴──────────┴──────────────────────────┐
│ API Layer (Node.js + Socket.io) │
│ ┌──────────────────────────────────────────────────────────┐ │
│ │ Real-Time Engine (Socket.io) — pushed to all clients │ │
│ └──────────────────────────────────────────────────────────┘ │
│ ┌──────────────────────────────────────────────────────────┐ │
│ │ Scan Pipeline (Market Trend → Sector → PDH/PDL → ...) │ │
│ └──────────────────────────────────────────────────────────┘ │
│ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ │ NSE Data │ │ Historical │ │ OI Data Aggregator │
│ │ Feed (WebSocket) │ Database │ │ (NSE F&O Bhavcopy) │
└──────────────────────────────────────────────────────────────┘
```

## Scanner Pipeline Details

### 7-Filter Pro Scanner

### AI Intraday Scanner — 7-Factor Scoring

Each of the 188 F&O stocks receives a score (0–100) based on weighted factors:

### 4-Phase Smart Scanner

### NIFTY OI Tracker

---

# Feature Deep-Dive — RICE Priority Matrix

| Feature | Reach (users/month) | Impact (1–5) | Confidence (%) | Effort (weeks) | RICE Score |
|---|---|---|---|---|---|
| 7-Filter Pro Scanner | 5,000 | 5.0 | 90% | 6 | 3,750 |
| Socket.io Real-Time Data | 5,000 | 5.0 | 95% | 4 | 5,937.5 |
| NIFTY OI Tracker + PCR | 3,000 | 4.5 | 85% | 3 | 3,825 |
| AI Intraday Scanner (7-factor) | 3,000 | 4.5 | 70% | 8 | 1,181.3 |
| 4-Phase Smart Scanner | 2,000 | 4.0 | 75% | 5 | 1,200 |
| Sector Heatmap | 2,500 | 3.5 | 80% | 3 | 2,333.3 |
| Real-Time Alerts (WebSocket) | 3,000 | 3.5 | 85% | 2 | 4,462.5 |
| Saved Scan Configurations | 1,500 | 3.0 | 80% | 1 | 3,600 |
| Export to CSV/PDF | 1,000 | 3.0 | 90% | 1 | 2,700 |
| Custom Formula Scanner | 500 | 4.0 | 50% | 6 | 166.7 |
| Watchlist Management | 2,000 | 2.5 | 85% | 1 | 4,250 |
| Mobile Responsive View | 500 | 3.0 | 60% | 3 | 300 |
| Dark Theme | 5,000 | 2.0 | 95% | 1 | 9,500 |

## Priority Ranking (Tiered)

**Tier 1 — Core Scanner (Weeks 1–6):** 1. Dark Theme (9,500) — Trivial; highly valued by traders 2. Socket.io Real-Time Data (5,937.5) — Foundation of the platform 3. Real-Time Alerts (4,462.5) — Critical for actionability 4. Watchlist Management (4,250) — Basic user workflow 5. NIFTY OI Tracker + PCR (3,825) — Key differentiator 6. 7-Filter Pro Scanner (3,750) — Core product

**Tier 2 — Intelligence (Weeks 7–12):** 7. Saved Scan Configurations (3,600) 8. Export to CSV/PDF (2,700) 9. Sector Heatmap (2,333.3) 10. 4-Phase Smart Scanner (1,200) 11. AI Intraday Scanner (1,181.3)

**Tier 3 — Advanced (Weeks 13–16):** 12. Mobile Responsive View (300) 13. Custom Formula Scanner (166.7)

---

# Success Metrics

| KPI | Baseline | Target (3 months) | Target (12 months) |
|---|---|---|---|
| Daily Active Users (DAU) | 0 | 500 | 5,000 |
| Scans executed/day | 0 | 2,500 | 50,000 |
| User retention (D7) | N/A | 40% | 60% |
| User retention (D30) | N/A | 20% | 35% |
| Paid conversion rate | N/A | 8% | 12% |
| Average session duration | N/A | 12 min | 20 min |
| Scanner accuracy (backtest of signals vs. actual moves) | N/A | 65% | 72% |
| NPS | N/A | 40 | 60 |

---

# Roadmap

## Phase 1 — Live Scanner Foundation (Weeks 1–6)

| Week | Deliverable |
|---|---|
| 1–2 | NSE real-time data feed integration (WebSocket); Socket.io server setup; basic stock list (188 F&O) |
| 3 | 7-Filter Pro Scanner pipeline (Filters 1–3: Market Trend, Sector, PDH/PDL) |
| 4 | Filters 4–7 (Volume/ATR, VWAP, RSI, OI); NIFTY OI Tracker with PCR |
| 5 | Real-time alerts engine; watchlist CRUD |
| 6 | Dark theme UI; scan results view; UAT with 10 beta traders |

## Phase 1 — Live Scanner Foundation (Weeks 1–6)

**Go/No-Go:** Scan refresh < 1 second; data latency < 500ms; 10 beta users active

## Phase 2 — AI Intelligence (Weeks 7–12)

**Go/No-Go:** AI scanner correlation ≥ 65% with next-15-min price direction

## Phase 3 — Scale & Monetise (Weeks 13–20)

---

# Impact Estimate

## Quantitative Impact

| Metric | Trader Without Tool | Trader With Tool | Value Created |
|---|---|---|---|
| Daily screening time | 45 minutes | 3 minutes | 42 min/day saved |
| Setups identified per day | 1–2 | 5–8 | 4× more opportunities |
| Win rate (self-reported) | ~55% | ~65% (estimated) | 10% improvement |
| Avg monthly turnover | ₹5 Lakh | ₹6.5 Lakh | 30% more |
| Missed opportunities (cost) | ~₹15,000/month | ~₹3,000/month | ₹12,000/month saved |

## Quantitative Impact

## Revenue Model

## Break-Even

- Development (4-person team, 20 weeks): **₹40 Lakh**
- Infrastructure (NSE data feed, Socket.io servers, Redis, DB): **₹5 Lakh/month**
- Customer acquisition (₹500/paid user × 8,000): **₹40 Lakh**
- Total Year 1 cost: **₹1.4 Cr**
- **Break-even: Month 16** (at 4,000 paid users)

- **Year 3 ARR target: ₹7.4 Cr**

---

*Document version: 1.0 | Next review: 2026-06-12*